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5 November 1973
SUPERSEDING
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10 January 1969

MILITARY SPECIFICATION

ASSAULT VEHICLE, FULL TRACKED,
AMPHIBIOUS: XM733

1. SCOPE

1.1 Scope. This specification covers one type of armored, full tracked, low ground pressure vehicle for cross country, amphibious, and phase I airborne operations.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION
Military

MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-G-3056	-	Gasoline, Automotive, Combat.
MIL-R-3065	-	Rubber, Fabricated Parts.
MIL-W-45205	-	Welding, Gas Metal-Arc and Gas Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor.
MIL-A-45564	-	Assault Vehicle, Full Tracked, Amphibious, XM733; Processing for Storage and Shipment of.
MIL-A-46100	-	Armor Plate, Steel, Wrought, High-Hardness.

FSC-2350

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MIL-E-46716 - Engine, Gasoline: 8-Cylinder, V-Type, Liquid-Cooled, 160 H.P.
MIL-T-46719 - Transmission, Hydraulic: Model 305MC.
MIL-S-46751 - Steer Unit with Vehicle Brakes and Final Drives, GS-100-3.
MIL-G-46795 - Generator System; Alternator, Rectifier 60 Amperes, 28 Volts.
MIL-W-62162 - Welding, High Hardness Armor.

STANDARDS

Federal

FED. STD. NO. 595 - Colors.

Military

MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-193 - Painting Procedures Tactical Vehicles (Tracked and Wheeled).
MIL-STD-210 - Climatic Extremes for Military Equipment.
MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests.)
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-882 - System Safety Program for System and Associated Subjects for Equipment, Requirements for.

PUBLICATIONS

Engineering Parts Lists

8736747 - Assault Vehicle, Full Tracked, Amphibious, XM733.

SUPPLEMENTARY QUALITY ASSURANCE PROVISIONS

Master List

8736747 - Master List of Supplementary Quality Assurance Provision for Assault Vehicle, XM733.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction vehicle. When specified (see 6.2), the complete preproduction vehicle proposed to be furnished under this specification shall be produced prior to the manufacture or fabrication of vehicles in quantity. The vehicle(s) when complete shall be submitted to the Government for examination

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and tests to determine conformance to the requirements of this specification. Vehicle(s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling under the supply contract.

3.2 First production vehicles. The vehicles selected, shall be fully representative of vehicles proposed to be furnished under the contract, shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.2.1 Reliability. The vehicle(s) Mean Miles Between Failure (MMBF) shall not be less than 640 miles during first 4000 miles of operation consisting of 25% hard surface roads (HSR), 25% secondary roads (SR) and 50% cross country (CC) operation. A failure for calculation of Mean Miles Between Failure (MMBF) shall be as defined by Joint CDC/AMC Wheel and Track Vehicle Mission Failure Definition, dated 27 March 1972. The definition (excluding the amplification paragraphs) reads:

a. A "failure" is defined as any malfunction of the end item that requires corrective action which cannot be deferred:
   (1) Until the next scheduled maintenance (exclusive of lubrication services) if organizational maintenance is prescribed for correction, or--
   (2) For the remainder of its specified life before overhaul, replacement, rebuild or salvage (as applicable) if direct general support maintenance is prescribed for correction.

b. Corrective action is not deferrable if the malfunction causes (or would cause if not corrected):
   (1) Inability to commence operation, cessation of operation or degradation of performance capability of the end item below that required for the accomplishment of its prescribed mission fuction, or--
   (2) A critical or catastrophic hazard to personnel or equipment as defined by MIL-STD-882.

c. Any malfunction which the operator or crew (user) can remedy will not be considered a failure provided that the repair is authorized or prescribed as an operator function and can be accomplished within 60 minutes using only controls, tools or spare parts incorporated in or carried with the end item.

3.2.2 Durability. The vehicle, with rated payload shall have a 61.3% probability of completing a minimum of 4000 miles without replacement, rebuild or overhaul of any of the major components, i.e., Engine, Transmission, Transfer Case, Differential, and shall have a 75% probability of completing the first 4000 miles without repair or rebuild of hull, when operated over terrain specified herein.

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3.2.3 Maintainability. The total scheduled maintenance excluding driver/crew checks and services shall not exceed 30 manhours and the total unscheduled maintenance shall not exceed 50 manhours during 4000 miles operation (25% HSR, 25% SR, 50% CC). The scheduled maintenance intervals shall not be less than 90 days or 750 miles which ever comes first. Maximum time to repair (to include diagnosis repair and verification) utilizing personnel normally employed at:

(1) Organization maintenance shall not exceed 8 hours 95% of the time.
(2) Direct support maintenance shall not exceed 12 hours 95% of the time.
(3) General support maintenance shall not exceed 20 hours 95% of the time in a backup direct support role.

3.3 Materials.

3.3.1 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL cited in 3.4, and necessary in the construction of this vehicle, the contractor shall be responsible for using only those products which are listed (by part or drawing number as QPL items) on the QPL in the referenced specification or which have been approved for inclusion on such lists. Manufacturer's inspection records shall specifically list such component(s), name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

3.3.2 Ozone resistant. Rubber products procured under requirements of MIL-R-3065 are to be ozone resistant to the degree specified by the C1 requirements of MIL-STD-417. Rubber components controlled by detail specifications shall be ozone resistant only when required by the specification referenced on the drawing.

3.4 Construction. Vehicle, components, sub-assemblies, and assemblies shall be fabricated and assembled into a complete vehicle in accordance with drawings listed or referred to in Engineering Parts Lists (EPL) 8736747. All parts, sub-assemblies, and assemblies requiring identification shall be identified in accordance with MIL-STD-130.

3.4.1 Sealing .

3.4.1.1 Sealer. The sealer used to seal vehicle shall prevent the entrance of water into the hull when vehicle is fording streams or rivers during cross country operation and when operating amphibiously.

3.4.1.2 Seals. When vehicle is fording or operating amphibiously, the seals shall restrict the entrance of foreign matter into bearings, which are exposed to contamination during these operations. All bearing seals shall restrict the leaking of lubricants from the bearings (see 6.5).

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3.4.1.3 Crew compartment seals. The crew compartment seals and components shall provide leakproof joints throughout the entire compartment.

3.4.2 Special kits. When specified (see 6.2) special kits shall be furnished and installed.

3.4.2.1 50 Cal. machine gun mount kit. The 50 cal. machine gun ring mount kit, when installed on vehicles shall be free of interference, misalinement and shall support the 50 cal. machine gun with ammunition and tray during 4,000 miles of endurance operations. The kit shall support the machine gun during the firing of 2,000 rounds of ammunition without failure and permanent set of any component.

3.4.2.2 81MM mortar kit. The 81MM mortar kit, when installed on vehicle shall be free of interference, misalinement and shall accept the 81MM mortar tube, baseplate, ammunition and associated mortar equipment. During the 4,000 miles of endurance operation and firing 2,000 rounds of ammunition and kit shall operate without failure and permanent set of any component.

3.4.3 Engine. The engine conforming to MIL-E-46716, shall operate throughout all rated speed ranges without leaking of the coolant, lubricants, fuel, and shall meet all functional requirements.

3.4.3.1 Engine air induction system. The air induction system, as installed, shall prevent entrance of all foreign matter.

3.4.4 Transmission. The transmission conforming to MIL-C-46719, shall operate throughout all gear and speed ranges without malfunction, and leaking of lubricants.

3.4.4.1 Steering unit. The steering unit conforming to MIL-S-46751, shall operate throughout all steering ranges without wander, side pull and leaking of lubricants.

3.4.5 Exhaust system. The engine exhaust system as installed shall be gas tight and leakproof.

3.4.6 Fuel system. The fuel system shall deliver an adequate supply of fuel to the engine when vehicle is ascending and descending longitudinal grades of 60 percent in forward and reverse gear, and operate on side slopes of 40 percent with each side of vehicle up slope.

3.4.6.1 Fuel tank. Provisions shall be made to assure the internal cleanliness of the fuel tank prior to initial fueling of vehicle. The fuel tank shall be capable of receiving 50 gallons of fuel per minute.

3.4.6.2 Fuel and hydraulic lines. Fuel and hydraulic lines and connections shall be internally clean prior to initial fueling of vehicle. All fuel and hydraulic line connections and tanks shall be free from leakage.

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3.4.6.3 Fuel shut off valve. When the manual fuel shut off valve is closed, the engine shall stop within five minutes.

3.4.6.4 Throttle linkage. With throttle linkage attached to engine, throttle pedal fully depressed, the throttle shall be in wide open position.

3.4.7 Electrical system.

3.4.7.1 Generating system voltage (alternator). The generator system shall conform to class B of MIL-G-46795. The generator system shall be adjusted prior to vehicle installation so when installed in vehicle, adjustment shall remain at 28 ± .5 volts under all conditions of vehicle operating conditions.

3.4.7.2 Lights. The internal and external lights shall operate under all vehicle operating conditions. All electrical contacts and connections shall maintain positive contact under all vehicle operating conditions.

3.4.7.3 Bilge pump motors. Bilge pump motors shall be capable of starting and operating satisfactorily on a minimum of 18.5 volts dc.

3.4.7.4 Communication and electrical circuits. The communication and electrical circuits shall be capable of maintaining continuity from end to end, without any shorts from pin to pin and from each pin to the connector shell. All electrical cables and wiring shall be anchored in such a manner to preclude any damage to the cables and wiring.

3.4.8 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction throughout all ranges of operation under all vehicle operating conditions.

3.4.9 Adjustment mechanisms. All adjustment mechanisms shall function properly, and maintain adjustment settings during all vehicle operating conditions.

3.4.10 Stowed material. All interior and exterior on vehicle material (OVM) items shall be stowed (applicable kits) on the vehicle in the spaces provided, to assure that items will not interfere with the operation of vehicle and components in any manner.

3.4.11 Air transportability. Vehicle combat loaded, shall weigh not more than 11,000 pounds (10,500 pounds for helicopter transport).

3.4.12 Welding and welding repairs (aluminum). Welding and welding repairs shall be in accordance with MIL-W-45205.

3.4.13 High hardness steel armor. When high hardness steel armor is used it shall meet requirements of MIL-A-46100, and welded as specified in MIL-W-62162.

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3.4.14 Riveting. The riveting shall be such that there shall be no leaking of water into the vehicles.

3.5 Break-in run.

3.5.1 Preparation. Prior to break-in run, except for equipment and component installation and servicing performed by the contractor as a final processing of vehicle for acceptance, the vehicle shall be completely assembled and all adjustments completed. Vehicle, including all components requiring lubrication shall be lubricated, and serviced with fuel conforming to MIL-G-3056. Lubricants and fluids shall be as specified in production lubrication drawing. Specific applications by grades or types for various expected seasonal temperature ranges shall be as specified.

3.5.2 Oil pressure and temperature. Prior to the start of the break-in run the engine shall be operated at 750 to 1000 rpm until lubricating oil is at operating pressure and temperature. If at any time during the break-in run the engine has been stopped for at least 15 minutes, the engine shall again be operated, as above, before continuing break-in run. Proper oil pressure and temperature shall be maintained during operation on level ground, 60 percent grades and 40 percent side slope.

3.5.3 Distance operated. Prior to tests, each vehicle shall be given a break-in run on smooth, level, hard-surfaced roads, except a section of the road shall be as shown in figure 1. The vehicle shall be operated as specified in table I and over the undulating section in a and b division of the break-in run in both directions for the purpose of observing the function and performance of various parts, assemblies, and controls.

Table I. Speeds and distances for break-in run

Division of run	Speed - MPH	Distance - Miles
a	0 to 10	10
b	11 to 20	10
c	21 to 30	10
d	31 to Max. Governed	10

3.5.3.1 Reverse operation. A ter each division of the break-in run, vehicle shall be stopped, the engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet.

3.5.4 Condition after run. After completion of the break-in run, there shall be no indication of damaged components maladjustment or loss of original adjustment that may cause faulty vehicle operation.

3.6 Performance. A complete vehicle, with applicable kits (see 3.4.2) and OVM (see 3.4.10) and serviced shall perform as specified herein. Vehicle, serviced (see 3.5.1) and equipped for prevailing climatic conditions, shall operate as specified with and without applicable kits. Performance shall be

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demonstrated on smooth, relatively level, hard-surfaced roads free of loose material, except a section of road shall be as shown in figure 1. The vehicle combat loaded or with a simulated load of equal weight shall be capable of operating on longitudinal grades up to 60 percent, and side slopes up to 40 percent with each side of vehicle up slope without stalling and damage to vehicle.

3.6.1 Environmental. The vehicle shall operate in ambient air temperature of plus 115° to minus 25°. The complete vehicle when in storage shall withstand climatic extremes as specified in MIL-STD-210, without deterioration that may cause failure of any component part of the vehicle.

3.6.2 Cooling system. The engine, transmission, and steer unit temperatures shall not exceed those shown in table II when the vehicle is operated under one or a combination of the following conditions:

(a) Ambient temperatures up to 115°F.
(b) Under full load, full throttle at 3.7 MPH vehicle speed.
(c) At specified speeds in water and maximum waves of 1 foot total amplitude.

Table II. Operating temperatures

Maximum engine coolant temp upstream of radiator	Maximum engine oil temp measured at gallery	Maximum transmission oil temp measured at sump	Maximum steer unit oil temp at sump
230°F	300°F	285°F	300°F

3.6.3 Bilge pumps. Each installed bilge pump shall remove water from the vehicle at a rate of not less than 44 gallons per minute (gpm) with a discharge head of three feet, and with pump motor operating at 27.5 volts impressed at the motor.

3.6.4 Vehicle speeds.

3.6.4.1 Speeds on land. The vehicle with applicable kits and OVE shall be capable of sustained speeds up to 30 miles per hour (mph), and shall be capable of maintaining a minimum speed of 1.5 mph in lowest gear range. Performance shall be demonstrated on smooth, relatively level, hard-surfaced roads. With vehicle traveling between 15 mph and maximum speeds vehicle drift shall not exceed 3 feet in 100 feet. The vehicle shall attain a maximum speed of 35 mph.

3.6.4.2 Speeds on grades. Vehicle shall operate at a sustained speed of 25 mph while ascending a 10 percent grade and at 5 mph while ascending a 60 percent grade.

3.6.5 Acceleration. With the vehicle at a stand-still position and engine at idle speed, the vehicle shall accelerate from 0 to 30 mph in not more than 15 seconds on smooth, relatively level, hard-surfaced roads.

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3.6.6 Engine starting on grades and side slopes. Vehicle standing on a 60 percent grade for not less than 2 minutes with engine operating under no load between 750 and 1000 rpm, followed by engine inoperative for not less than 2 minutes, the engine shall restart in not more than 1 minute with vehicle headed up and down longitudinal grade. With vehicle standing on a 40 percent side slope, the engine shall restart in not more than 1 minute, under the same conditions specified for starting on grades, with right and left side of vehicle up slope.

3.6.7 Braking.

3.6.7.1 Stopping. Vehicle traveling at 20 mph, on dry, smooth, relatively level, hard-surfaced roads free of loose material shall be stopped by service brake within a distance of 30 feet from point of brake application. Vehicle drift shall not exceed 3 feet in stopping distance.

3.6.7.2 Holding. With vehicle on a 60 percent grade, transmission in neutral, and brake levers held or brakes locked for parking, the vehicle shall be held stationary for not less than 2 minutes. Vehicle shall be combat loaded, or with a simulated load of equal weight.

3.6.8 Turning. With the vehicle in geared steer, operating on smooth, relatively level, hard-surfaced road, shall make continuous full 360 degree turns, to right and to left, with rear point of outer edge of vehicle at all times in contact with or within a 40 foot radius. When pivot steer is employed the vehicle shall pivot within a 15 foot radius.

3.6.9 Fording and amphibious operation.

3.6.9.1 Shallow water operation. Vehicle shall be capable of fording hard-bottom salt or fresh water up to the depth required to float vehicle, without damage to vehicle and components.

3.6.9.2 Amphibious operation. Vehicle, combat loaded or with a simulated load of equal weight and launched amphibiously, shall operate in water of unlimited depth and waves not exceeding 1 foot total amplitude, and water current not greater than 4 miles per hour. When afloat, in calm water of not less than 28 feet in depth, the vehicle shall attain a maximum speed of not less than 3.5 mph. The vehicle, when crossing calm, open water shall operate within a 30 foot corridor.

3.6.9.3 Leakage. During fording and amphibious operations the accumulation of water shall be not more than 1 inch in the hull when operating in water for 30 minutes. Following fording and amphibious operation, the contamination of lubricants shall be not more than 2 percent.

3.6.10 Trench crossing. Vehicle shall cross trenches of unlimited depth and up to 60 inches in width.

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3.6.11 Vertical obstacles. Vehicle shall cross over obstacles up to 18 inches high when operated in forward gear and up to 12 inches in reverse gear.

3.6.12 Climbing. Vehicle shall ascend a prepared 60 percent longitudinal grade in forward and reverse gear from a stand-still position without stalling or damage to power train and power plant.

3.6.13 Winch. The winch shall provide a minimum line pull of 5,000 pounds on a bare drum, with engine operating at 750 to 850 rpm and with transmission in low gear. A safety device shall be provided to preclude damage to winch if line pull should exceed 120 percent of the rated capacity.

3.6.14 Vibrations and noise level.

3.6.14.1 Equipment operation. Equipment operation for training or maintenance purposes shall not exceed the maximum acceptable noise level specified in table III. Noise reduction principles and techniques applied to the equipment shall be determined by the procuring activity. Should the cost of reducing the sound pressure levels to meet the maximum of or below the requirements of table III become prohibitive, it shall be necessary to obtain the approval of the procuring activity before proceeding with the reduction.

3.6.14.2 System effectiveness. The reduction of the sound pressure level to or below the maximum acceptable noise level of Table III shall not degrade the system effectiveness.

Table III. Maximum Acceptable Noise Level for Army Materiel Command Equipment (Continuous Noise)

Frequency bands, CPS	Maximum acceptable noise level (db re 0.0002 microbar)
37.5 - 75	120
75 - 150	115
150 - 300	109
300 - 600	101
600 - 1200	93
1200 - 2400	89
2400 - 4800	89
4800 - 9600	91

3.6.14.3 Distance from noise source. The distance from the noise source at which the sound pressure level is equal to or below that specified in the table III shall be determined.

3.6.14.3.1 Operator's protection. With the equipment operating and the sound pressure level of the item of equipment exceeds the maximum acceptable noise levels of the table, the equipment shall be conspiciously marked as follows: "Warning, ear protection required within the specified distance when equipment is operating".

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3.6.15 Electromagnetic compatibility. Unless otherwise specified (see 6.8), the vehicle shall be radio interference suppressed in accordance with the tactical vehicle requirements of Class IIIA of MIL-STD-461.

3.7 Fire extinguishers.

3.7.1 Portable. The portable fire extinguishers, when mounted shall be readily accessible for servicing and use.

3.7.2 Fixed automatic fire extinguisher detection system. The automatic fire extinguisher detection suppression system shall be installed in a manner that shall assure protection of the electrical wiring controls and suppression lines from chafing and abrasions during all conditions of vehicle operation.

3.8 Painting, marking, and data plates.

3.8.1 Painting. Exterior and interior of vehicle and component assemblies and parts which require painting, shall be prepared and painted in accordance with MIL-STD-193. Color shall be in accordance with FED. STD. NO. 595, as specified for applicable service (see 6.2).

3.8.2 Marking. Vehicle marking shall be in accordance with MIL-STD-642, as specified for applicable service (see 6.2).

3.8.3 Name, shipping, and service data plates. Data plates shall be in accordance with MIL-P-514.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor quality assurance system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective inspection or quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent limitation is included in the approved quality assurance procedure. The contractor shall notify the Government and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents.

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4.1.2 Parts and components. Parts, components and assemblies shall be inspected in accordance with the drawings listed on EPL 8736747 and Master List of Supplementary Quality Assurance Provision 8736747.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Qualified products. The contractor's inspection records shall be checked to determine contractors conformance to 3.3.1.

4.2 Preproduction and first production inspection.

4.2.1 Preproduction inspection. A preproduction vehicle (see 3.1) when furnished shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed at a location approved by the Government. Failure of the preproduction model to comply with any of the requirements specified shall be cause for refusal by the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.2.2 First production inspection.

4.2.2.1 In-process inspection. During fabrication of the first production vehicle (see 3.2) an in-process inspection will be conducted by USATACOM. Product Assurance representatives to evaluate conformance of materials and workmanship to requirements of specified documents. The chassis, suspension and electrical components shall be available for inspection. These inspections shall be made at the contractor or subcontractor's facility prior to application of primer and paint. Processing and welding procedures, quality system and inspection records will be reviewed and evaluated during this inspection.

4.2.2.2 Completed first vehicle inspection. The first production vehicle of each model on each contract shall be road tested and completely inspected by the contractor for conformance to the requirements of contract and specfications. The applicable road test shall be in accordance with paragraph 4.5.1. Upon completion, the contractor shall submit the vehicle (and make available all inspection records and certifications) to the responsible government inspection element at contractor's plant for preliminary inspection.

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4.2.2.2.1 Preliminary inspection. The vehicle shall receive a preliminary inspection by the responsible Government inspection element. The preliminary inspection shall be complete except for road test requirements of paragraph 4.5.1 which will be conducted jointly during provisional inspection to be accomplished by representatives of TACOM Product Assurance Directorate as specified in paragraph 4.2.2.2.2.

4.2.2.2.2 Provisional inspection. The first completed production vehicle shall be subject to provisional inspection at the contractor's plant by representatives of TACOM Product Assurance Directorate and shall further provide any other inspection assistance as may be required. At the time of this inspection, the contractor shall make available his inspection plan, inspection records and certifications pertinent to the vehicle and components.

4.2.2.2.3 Corrections. Corrections found necessary as a result of above inspections shall be made by the contractor.

4.2.2.2.4 Vehicle disposition. After completion of inspection, the first production vehicle of a specific model shall remain at the manufacturing facility, as a production sample, and be the last vehicle shipped on the contract. The chassis may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the vehicle during this period in accordance with applicable documents for care and preservation while in storage.

4.2.2.3 Final approval and acceptance. Final approval and acceptance by the Government of the first vehicle of a specific model shall be withheld until the 4000 mile test of the second vehicle produced has been completed and a final determination has been made regarding conformity of the vehicle to contractual requirements, including, but not limited to workmanship and materials.

4.2.3 Production vehicle test. After completion of the first vehicle inspection, a second vehicle shall be selected from the first months production or one of the first 10 shall be subjected to examination (see 4.4.2) and tests, specified in 4.5 (a) and (b). Subsequent to examination and tests, the selected vehicle shall be subjected to the 4,000 mile endurance and performance tests specified in table IV at a site approved by the Government, which will require a maximum of 90 days. These tests are to be performed by the Government. Delay in test completion caused by vehicle breakdown or failure of the contractor to adequately support the vehicle with parts during the tests shall not be the basis for adjustment of the contract, delivery schedule or contract price.

Table IV. 4,000 mile test combat loaded (Government Proving Grounds)

Course	Mileage and speeds
Hard surface roads	1,000 miles at varying speeds up to max.
Gravel or dirt roads	1,000 miles at varying speeds up to max.
Level cross country	1,000 miles at varying speeds up to max.
Hilly cross country	1,000 miles at varying speeds up to max.
Amphibious operation	1 hour minimum at varyng speeds up to maximum

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4.2.3.1 Reliability verification. Reliability requirements of 3.2.1 will be verified at a 50% confidence level while vehicle is subjected to initial production tests. Mean Miles Between Mission Failure (MMBF) shall not be less than specified.

4.2.3.2 Durability verification. To determine conformance to 3.2.2, the durability requirement will be verified at a 50% confidence level while the vehicle is subjected to initial production test.

4.2.3.3 Maintainability verification. The maintainability of the vehicle as specified in 3.2.3 will be verified during initial production testing.

4.2.4 Maintainability failures. Any failures or deficiencies of a workmanship or materials nature found during or as a result of the 4,000 test, will be cause for rejection of the vehicle. Further, the Government will refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of the 4,000 mile test shall be prima facie evidence that all vehicles already accepted prior to completion of the 4,000 mile test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 Break-in run (track). To determine conformance to 3.5.1, 3.5.2, and 3.5.3 each vehicle shall be lubricated and serviced prior to the break-in run. The run shall be on smooth, level, hard-surface roads, and for the distance specified in table I. The vehicle shall be operated in reverse after each division as specified in 3.5.3.1. The undulating portion of the test shall be accomplished twice in each direction at speeds between 5 and 10 mph during division "a" and between 11 and 14.5 mph during division "b".

4.3.1 Condition after run. After the break-in run the vehicle shall be inspected to determine conformance to 3.5.4.

4.4 Inspection provisions.

4.4.1 Lot size. An inspection lot shall consist of all vehicles of one type, from one day's production submitted at one time for acceptance inspection.

4.4.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected, from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling may be accomplished by the contractor, the contractor shall 100 percent inspect the first 20 vehicles to establish a process average (see 6.4) to allow normal sampling in accordance with MIL-STD-105.

4.4.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table V. The following constitutes a part of the classification of defects (see 6.5):

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(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any leak that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

Table V. Classification of Defects

Characteristic	Major = AQL 25 Def./100 Units	Minor = AQL 150 Def./100 Units	Method of Inspection
Engine	101. Malfunction leaks (see 6.5), b mounting	201. Improper component assy.	Visual & functional
Transmission	102. Malfunction, leaks (see 6.5), b mounting improper	202. Improper component assy or installation	Visual & functional
Steer unit	103. Malfunction, leaks (see 6.5), a	203. Improper component assy.	Visual & functional
*Fuel system components	104. Malfunction, leaks (see 6.5), a	204. Improper line clearance, assy. or installation	Visual & functional
Cooling system components	105. Malfunction, damage leaks (see 6.5) b	205. Improper assy., or installation	Visual & functional
Engine governor	106. Malfunction, seals, defective or missing		Visual & functional
Instrumentation switches, warning, indicating and safety devices	107. Malfunction, damage, calibration	206. Improper assy. or installation calibration	Visual & functional
Ignition system components	108. Malfunction, damage	207. Improper assy., or installation	Visual & functional
Electrical system components and circuits, communication system, **lights	109. Malfunction, lighting sequence improper	208. Improper assy., or installation	Visual & functional

*Fuel leaks constitute a special defect and will result in total inspection of the lot for this deficiency.

**Filament failures are not to be classified as a major or minor defect.

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Table V. Classification of Defects (Cont'd)

Characteristic	Major = AQL 25 Def./100 Units	Minor = AQL 150 Def./100 Units	Method of Inspection
Suspension system components adjustments	110. Damage, leaks (see 6.5) b clearance improper misalignment	209. Improper installation or assy., bolt or nut torque	Visual & functional
Tracks	111. Damage, pad separation, guide welding defects	210. Tension, installation or assy., improper, including bolt torque	Visual
Doors, covers, grilles, ducts, vents, panels, stowage boxes, brackets, stops, seats, fenders, hardware	112. Malfunction, torsion spring, hold-open, locking devices, damage	211. Improper fits, adjustment, installation or assy. defective welds, seals, adhesion, hardware or sheet metal	Visual & functional
Controls	113. Malfunction, clearance improper	212. Improper assy., or installation	Visual & functional
Track drive sprockets & drive shafts	114. Damage, clearance improper	213. Improper assy. installation or bolt torque	Visual
Adjustment mechanisms	115. Malfunction, clearance improper	214. Improper assy., or installation	Visual & functional
Hull, riveting	116. Structural, leaking at rivets & welding defects		Visual
Towing pintle, lifting eyes, hooks & brackets	117. Malfunction or welding defects	215. Improper assy., or installation	Visual & functional
Seats & straps	118. Damages, seat control malfunction	216. Improper fit, installation, assy., or adhesion, welding defects	Visual & functional

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Table V. Classification of Defects (Cont'd)

Characteristic	Major = AQL 25 Def./100 Units	Minor = AQL 150 Def./100 Units	Method of Inspection
Fire extinguisher	119. Damage	217. Improper assy., or installation	Visual
Wiring, tubing, tube fittings, and hoses	120. Defective, or improperly secured	218. Improper assy., installation, coding or protection	Visual
Paint		219. Application & color improper	Visual
Decals, marking data and instruction plates		220. Missing, improper location or sizes, defective	Visual
Lubrication & lubricants		221. Improper application, types & levels	Visual
Protective & anti-skid coatings		222. Application, materials & coverage improper	Visual
Record forms & publications		223. Missing, improperly processed or secured	Visual

4.4.2.1 Unclassified defects. All defects having no effect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty-day period will be added to the minor classification with no increase in AQL's.

4.4.2.2 Recurring major deficiencies. A major deficiency is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found. Applicable only when inspection is performed on a sampling basis.

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4.4.2.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. Correction of all deficiencies shall be accomplished prior to acceptance. Applicable only when inspection is performed on a sampling basis.

4.5 Classification of tests.

(a) Acceptance tests 4.5.1 through 4.5.1.23.
(b) Control tests 4.5.2 through 4.5.2.9.
(c) Comparison test (CT) 4.5.3 through 4.5.3.13.

4.5.1 Acceptance tests. To determine conformance to 3.6 after break-in run (see 3.5) each vehicle shall be operated for a distance of eight miles or more by the contractor and subjected to acceptance tests as specified in Table VI. Vehicles shall be completely assembled and serviced (see 3.5.1) with fuel tank filled to capacity, lubricants, and fluids as specified, but need not be combat loaded, except for holding and amphibious operational tests as specified in 4.5.1.18 and 4.5.2.8. After the tests, the vehicle shall be examined for evidence of fuel or lubricant leakage, or other deficiencies.

4.5.1.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government inspector shall stop acceptance until evidence has been provided by the contractor that corrective action has been taken.

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Table VI. Location for tests

Title	Paragraph Reference Numbers	Place of Manufacture	Government Proving Ground
Acceptance Tests	4.5.1	X	
Sealing check	4.5.1.2.1	X	
Seals check	4.5.1.2.2	X	X
Engine check	4.5.1.3	X	X
Engine air induction system check	4.5.1.4	X	X
Transmission test	4.5.1.5	X	X
Exhaust system check	4.5.1.6	X	X
Fuel and hydraulic system leakage check	4.5.1.7	X	X
Fuel shut off valve check	4.5.1.8	X	X
Throttle linkage check	4.5.1.9	X	X
Generating system voltage check	4.5.1.10.1	X	X
Light check	4.5.1.10.2	X	X
Bilge pump electrical motor check	4.5.1.10.3	X	X
Communication and electrical equipment test	4.5.1.10.4	X	X
Controls check	4.5.1.11	X	X
Adjustment mechanism check	4.5.1.12	X	X
Welding & welding repair check	4.5.1.13	X	X
Speed test	4.5.1.14	X	X
Acceleration test	4.5.1.15	X	X
Engine starting on grades and side slope test	4.5.1.16	X	X
Stopping test	4.5.1.17	X	X
Holding test	4.5.1.18	X	X
Turning & steering unit test	4.5.1.19	X	X
Shallow water fording test	4.5.1.20	X	X
Winch check	4.5.1.21	X	X
Vibration & noise test	4.5.1.22 thru 4.5.1.22.2	X	X
Fire extinguisher systems check	4.5.1.23	X	X
Control Tests	4.5.2		
Special kit tests	4.5.2.3	X	X
Fuel tank check	4.5.2.4	X	X
Communication equipment check	4.5.2.5	X	X
Stowed equipment check	4.5.2.6	X	X
Bilge pump operational check	4.5.2.7	X	X
Amphibious operational test	4.5.2.8	X	X
Fixed fire extinguisher system test	4.5.2.9	X	X

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Table VI. Location for tests (Cont'd)

Title	Paragraph Reference Numbers	Place of Manufacture	Government Proving Ground
Comparison Test	4.5.3		X
Fuel system slope test	4.5.3.2		X
Air transportability check	4.5.3.3		X
Longitudinal grade, side slope operational and speed on grades test	4.5.3.4		X
Environmental tests	4.5.3.5		X
Cooling system test	4.5.3.6		X
Water removal bilge pump check	4.5.3.7		X
Amphibious operational test	4.5.3.8		X
Trench crossing test	4.5.3.9		X
Vertical obstacle test	4.5.3.10		X
Climbing test	4.5.3.11		X
Radio interference suppression test	4.5.3.12		X
Fixed fire extinguisher detection system test	4.5.3.13		X

4.5.1.2 Sealing and seals check.

4.5.1.2.1 Sealing check. To determine conformance to 3.4.1.1, prior to and after fording and amphibious operations the vehicle shall be checked for water in vehicle, leaking of lubricants, and crew compartment leaking.

4.5.1.2.2 Seals check. To determine conformance to 3.4.1.2 and 3.4.1.3, prior to and after fording and amphibious operations the bearing seals shall be checked for lubricant leaks, and leaks into the crew compartment.

4.5.1.3 Engine check. To determine conformance to 3.4.3, the engine shall be checked for proper installation, any leaking of lubricants, fuel, and coolants.

4.5.1.4 Engine air induction system check. To determine conformance to 3.4.3.1, the engine air induction system shall be checked for proper installation and leaks.

4.5.1.5 Transmission test. To determine conformance to 3.4.4 the transmission shall be operated throughout all gear and speed ranges and steering requirements and check for performance requirements.

4.5.1.6 Exhaust system check. To determine conformance to 3.4.5, the exhaust system shall be checked for proper installation and leaks.

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4.5.1.7 Fuel and hydraulic system leakage check. To determine conformance to 3.4.6 through 3.4.6.2, the fuel and hydraulic system shall be checked for proper installation and leakage. Fuel tank test shall include filling tank to capacity.

4.5.1.8 Fuel shut off valve check. To determine conformance to 3.4.6.3, with the engine operating at 750 to 900 rpm, the fuel shut-off valve shall be closed. The time for the engine to stop operation shall be as specified.

4.5.1.9 Throttle linkage check. To determine conformance to 3.4.6.4, the throttle pedal shall be operated to full throttle, and the engine throttle rod checked to assure that full throttle position has been reached when pedal is in contact with stop.

4.5.1.10 Electrical system tests.

4.5.1.10.1 Generating system voltage check (alternator). To determine conformance to 3.4.7.1, with the alternator operating as specified, a final voltage check shall be made, with a voltmeter, during or immediately after the road test.

4.5.1.10.2 Light check. To determine conformance to 3.4.7.2, all lights shall be operated during and after break-in run throughout all stages of operation, and checked for functional requirements.

4.5.1.10.3 Bilge pump electric motor check. To determine conformance to 3.4.7.3, the heater and bilge pump electrical controls shall be turned on and the motors checked for operation.

4.5.1.10.4 Communication and electrical equipment test. To determine conformance to 3.4.7.4, the electrical circuits supplying current to the communication and electrical equipment shall be functionally checked at all points to assure continuity throughout the circuits.

4.5.1.11 Controls check. To determine conformance to 3.4.8, all controls shall be operated and checked for functional requirements.

4.5.1.12 Adjustment mechanism check. To determine conformance to 3.4.9, all adjustment mechanisms shall be checked for proper adjustment and adjusted if required.

4.5.1.13 Welding, welding repairs and riveting check. To determine conformance to 3.4.12, 3.4.13, and 3.4.14, all welding and welding repairs and riveting repairs shall be checked for proper use of procedures and application.

4.5.1.14 Speed on land test. To determine conformance to 3.6.4.1, the vehicle shall be operated at specified speeds and checked for performance requirements.

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4.5.1.15 Acceleration test. To determine conformance to 3.6.5, the vehicle shall be accelerated from a standing start to the required miles per hour and the time checked.

4.5.1.6 Engine starting on grades and side slopes test. To determine conformance to 3.5.2, and 3.6.6, the engine shall start, oil pressure and temperature shall be maintained when operating on specified grades and slopes. The 60 percent grade test may be conducted in conjunction with brake holding test.

4.5.1.17 Stopping test. To determine conformance to 3.6.7.1, the vehicle shall be operated on dry level hard-surfaced road free of loose material, and the results of 3 stopping tests shall be averaged to make this determination.

4.5.1.18 Holding test. To determine conformance to 3.6.7.2, the vehicle shall be driven up specified grade, brakes applied and vehicle observed for movement, then with brakes locked for parking the vehicle shall be observed for movement. During this test vehicle shall be combat loaded or with a simulated load of equal weight shall be placed on the vehicle, in a position that will not restrict the flow of engine air (intake or exhaust).

4.5.1.19 Steering and turning tests. To determine conformance to 3.4.4.1 and 3.6.8, vehicle shall be operated and the steering unit checked for proper functional requirements. The vehicle shall be turned to right and left in full 360° using gear steer and then pivot turn, and checked for diameter of turns.

4.5.1.20 Shallow water fording test. To determine conformance to 3.4.14, 3.6.9.1, and 3.6.9.3 the vehicle shall be driven into water of specified depth without special equipment. Water accumulation shall be not more than specified quantity, in hull bottom, during 15 minutes of fording operation.

4.5.1.21 Winch check. To determine conformance to 3.6.13, the winch controls shall be operated as specified and winch observed for operating requirements.

4.5.1.22 Vibration and noise level check. To determine conformance to 3.6.14 through 3.6.14.3.1 the average of 3 noise level readings at each octave band shall be determined for the following situations and combinations.

(a) At each operation head position under normal operational procedures.
(b) At each instruction and trainee head position where training is conducted.
(c) At each maintenance personnel head position, where maintenance is conducted with the equipment operating.
(d) All positions where other personnel may be present in or on the equipment, representatives sample readings shall be made at the position occupied by these personnel as specified above.

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4.5.1.22.1 Sound level meter and microphone. The sound level meter and microphone used shall be in accordance with applicable documents (see 6.7).

4.5.1.22.2 Tones or narrow band noise. To determine conformance to 3.6. 14.2 Table III, when pure tones, or narrow bands of noise are present or any octave band, the sound pressure level of that octave band shall be reduced from the level shown in the table by 5db for frequencies above 1,000 cps and 10db for the frequencies below 1,000 cps.

4.5.1.23 Fire extinguisher systems check. To determine conformance to 3.7 through 3.7.2, the fire extinguishing systems shall be check to assure that all equipment and components required are as specified and function as required without interference.

4.5.2 Control test. Control tests for maintaining control of those requirements whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein.

4.5.2.1 Frequency. The first vehicle off the production line, or one of the first ten vehicles produced, and thereafter one per month, shall be checked in accordance with the following control tests except test specified in 4.5.2.4 shall be accomplished on 1 vehicle from each 10 produced.

4.5.2.2 Test failure. If a vehicle fails to pass any control test specified herein, the Government inspector shall stop acceptance of subsequent lots until evidence has been produced by the contractor that corrective action has been taken.

4.5.2.3 Special kit tests. To determine conformance to 3.4.2 through 3.4.2.2, each selected vehicle with kit installed shall be examined to assure the kit is complete and meets performance requirements.

4.5.2.4 Fuel tank check. To determine conformance to 3.4.6.1, the fuel tank shall be filled to capacity at a rate of not less than 50 gallons per minute. Before and after road test on smooth, level, hard-surfaced roads the tank and lines shall be checked for leaks.

4.5.2.5 Communication equipment check. To determine conformance to 3.4.7.4, the communication equipment leads and cable assemblies shall be checked for installation and functional requirements.

4.5.2.6 Stowed equipment check. To determine conformance to 3.4.10, all kits and depot installed OVE shall be stowed on the selected vehicle. All interior and exterior items shall fit in the spaces provided. The equipment shall be removed after the test. The OVE, used shall be of the latest production available to the contractor. The contractor shall be permitted to unpackage and repackage equipment used for test.

4.5.2.7 Bilge pump operational check. To determine conformance to 3.6.3, the bilge pumps shall be operated and checked for functional requirements.

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4.5.2.8 Amphibious flotation test. To determine conformance to 3.6.9.3, the vehicles shall be launched amphibiously and observed for flotation requirements.

4.5.2.9 Fixed fire extinguisher system test. To determine conformance to 3.7.2, the fixed fire extinguisher system shall be operated as required to assure that all electrical and other components are meeting performance requirements.

4.5.3 Comparison test. The Government may select vehicles at any time during the contract production period and subject vehicles to applicable tests specified in Table VI and Table VII (Government Proving Ground tests). Tests shall be conducted by the Government at a site selected or approved by the Government. Vehicles selected for proving ground tests shall not include any vehicles previously tested for conformance to 4.5.2 by the manufacturer.

4.5.3.1 Test failure. Failure of any vehicle tested to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during or as a result of the test cycle; may be considered cause for refusal to continue acceptance of vehicles by the Government until objective evidence has been provided by the manufacturer that corrective action has been taken to eliminate the condition which caused the failure.

Table VII. 2000 mile test combat loaded (Government Proving Ground)

Course	Mileage & Speeds
Hard surface or gravel	200 miles at varying speeds up to max.
Gravel and dirt road	500 miles at varying speeds up to max.
Level cross country	650 miles at varying speeds up to max.
Hilly cross country	650 miles at varying speeds up to max.
Amphibious	One (1) hour minimum at varying speeds up to max.

4.5.3.2 Fuel system slope test. To determine conformance to 3.4.6, with the vehicle operating on specified slopes the vehicle fuel system shall be observed for functional requirements.

4.5.3.3 Air transportability check. To determine conformance to 3.4.11, the vehicle shall be loaded as specified and weight checked.

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4.5.3.4 Longitudinal grade, side slope operational, and speed on grades test. To determine conformance to 3.6 and 3.6.4.2, the vehicle shall be operated on specified grades and slopes, and vehicle checked for performance requirements.

4.5.3.5 Environmental tests. To determine conformance to 3.6 and 3.6.1, vehicle and equipment shall be operated in specified temperatures and observed for performance requirements.

4.5.3.6 Cooling system test. To determine conformance to 3.6.2, the vehicle shall be operated in specified temperature and cooling system checked for functional requirements.

4.5.3.7 Water removal bilge pump check. To determine conformance to 3.6.3, the vehicle shall have sufficient water in hull, operate bilge pump and check the rate of water removal with the specified discharge head.

4.5.3.8 Amphibious operational test. To determine conformance to 3.4.14, 3.6.9.2, and 3.6.9.3, the vehicle shall be operated amphibiously and observed for performance requirements. Water accumulation shall be not more than specified quantity in hull bottom during 30 minutes of amphibious operation.

4.5.3.9 Trench crossing test. To determine conformance to 3.6.10, the vehicle shall be operated over trenches of specified depth and width, and vehicle checked for damage.

4.5.3.10 Vertical obstacle test. To determine conformance to 3.6.11, the vehicle shall be operated over vertical obstacles of specified height, and vehicle checked for damage.

4.5.3.11 Climbing test. To determine conformance to 3.6.12, the vehicle shall be given a climbing test on specified grade, and checked for performance requirements.

4.5.3.12 Electromagnetic compatibility test. To determine conformance to 3.6.15, the vehicle shall be subjected to the electromagnetic compatibility test as specified.

4.5.3.13 Fixed fire extinguisher detection system test. To determine conformance to 3.7.2, the fixed fire extinguisher detection system shall be operated as required to assure that all detection and extinguishing components meet their performance requirements.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and storage shall be in accordance with MIL-S-45564.

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6. NOTES

6.1 Intended use. This vehicle is intended for use by the Armed Forces in climatic zones, temperate, and tropical, as a weapon, and a personnel carrier.

6.2 Procurement document should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction model if required (see 3.1).
(c) Special kit if required (see 3.4.2).
(d) Color as required by using service (see 3.8.1).
(e) Marking as required by using service (see 3.8.2).
(f) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.3 Safety precautions. Caution should be used in handling carbon dioxide fire-extinguisher cylinders. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control since physical injury is highly probable.

6.4 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the AQL specifid in the classification of defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process Average} = \frac{\text{Number of Defects}}{\text{Number of Vehicles Inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100% inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

6.5 The following shall be used with the classification of defects for leaks:

a. Weep: Any evidence of fluid beyond the seal.
b. Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.
c. Droplet: Any evidence of fluid beyond the seal that results in formation of a droplet.
d. Drip: Any evidence of fluid beyond the seal where droplet forms and falls.

6.6 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the

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deficiency Sheet for the vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

6.7 The following publications should be used in determining sound pressure level.

ASA Z24.10-1953	Octave-Band Filter Set for the Analysis of Noise and Other Sounds
ASA S1.1-1960	Acoustical Terminology
ASA S1.4-1961	General Purpose Sound Level Meters

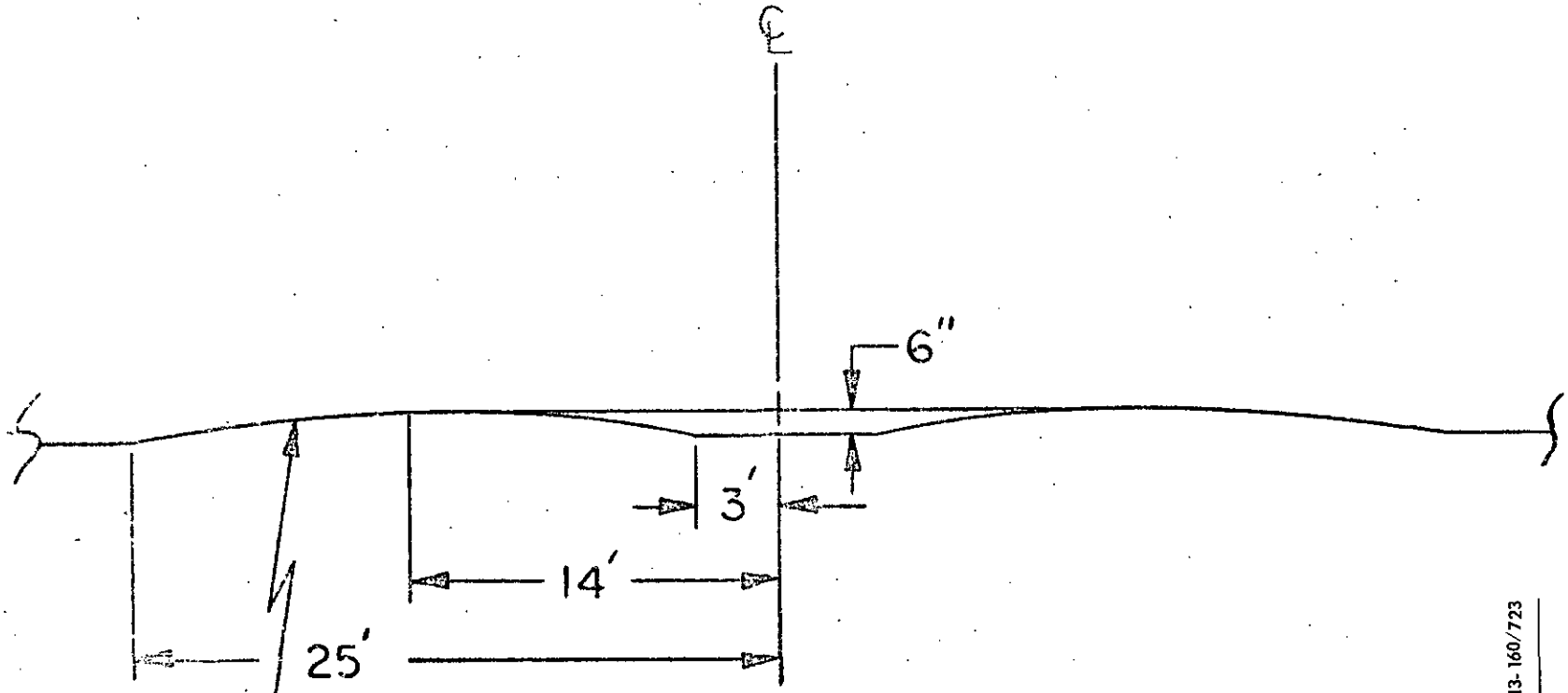
6.8 Electromagnetic compatibility test. The contracting officer should submit a copy of the report(s) required in 4.5.3.12 to the Commanding General U.S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey 07703, for approval.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2350-A241

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PROFILE OF TEST TRACK
(ARTIFICIAL BUMP)

FIG 1

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